An Automated Platform for GNSS Processing and Time-Series Analysis in a Complex Geodynamic Region

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Presentation Structure

- GNSS Networks in Greece
- Processing
- Results & Outputs
- Discussion / Conclusions

Motivation

Routine GNSS processing and site/network monitoring is crucial, because:

- Greece lies in a region of utmost tectonic and volcanic unrest (e.g. active volcano in Santorini isl.),
- results & products are important to a series of fields spanning the whole range of Geosciences,
- helps us follow and apply state-of-the-art technologies in GNSS analysis & Satellite Geodesy and expand & modernize our research activity,
- contribute to the GNSS/EUREF community and be involved in ongoing/future projects,
- develop new features for high rate processing.
- improve our academic services (NTUA is a University)

Throughout the last years, routine processing & monitoring has hepled us gain a more thorough view of the complex tectonic and volcanic setting of Greece.

The Data Set

Routine processing for precise positioning, assumes a well established, credible dataset (metadata).

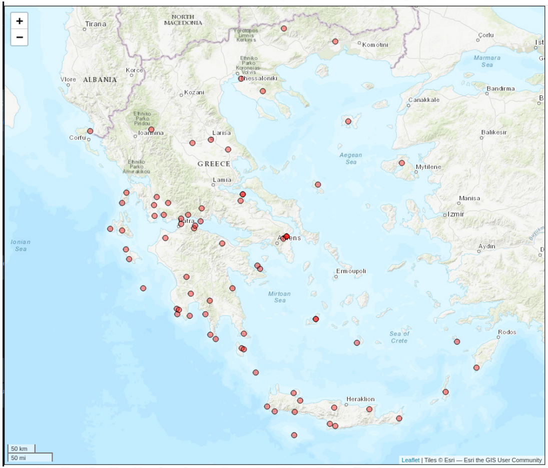
Currently we process whatever we can get our hands on ...
Problems:

- Inhomogenous dataset (`RINEX` of various versions, raw files, etc).
- Various maintainers, different mentalities.
- Different aquisition methods/rates.
- No log files for mainteners with no geodetic interest (e.g. surveying companies).
- Wide variety of equipment (not always included in `atx` files).

Network GREECE

Network **Greece** includes the majority of the available sites (≈100) but not all of them are (always/currently) active. Various providers but all with geodetic interest & equipment.

- covers all of Greece
- different (geodetic type) equipment
- credible time-span (early 2004 - now)
- all free available GNSS data
- large data gaps & inactive stations

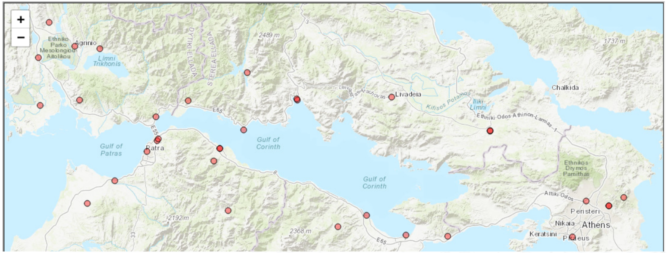


-» Network GREECE

Network Santorini

The **`Santorini Volcano`** ...

- credible time-span
- only covers the Corinth Rift
- different providers (including surveying & cadastral services)
- no log files & equipment changes



-» Network EnCeladus

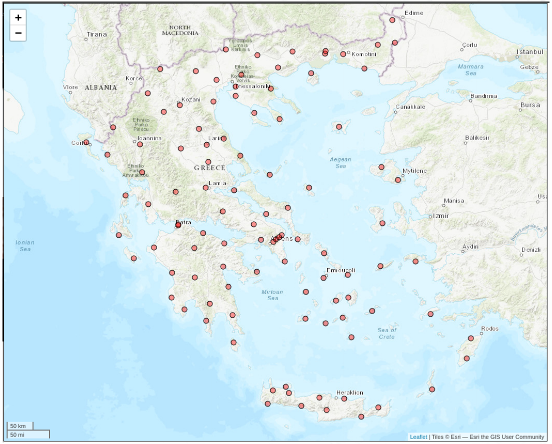
Network HEPOS

Network HEPOS

- covers all of Greece
- homogeneous equipment
- credible time-span (mid 2007 - now)
- only four years available for processing up to now

Data availability and funding from Hellenic Cadastre

HELLENIC CADASTRE



-» Network HEPOS

GNSS data analysis at DSO

GNSS data analysis

- Bernese v5.4 (**Bernese**)
- PRIDE-PPPAR (**PRIDE**)
- GipsyX (**GipsyX**)

Time series analysis

- Hector v2.1 (**Bos2012**)
- Python/C++ routines

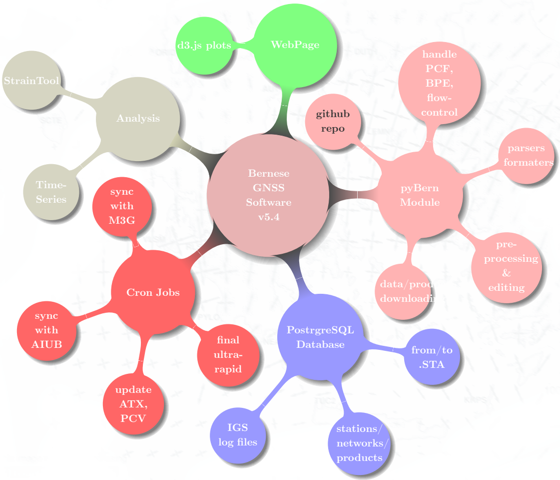
Solutions

- Daily solution
  - Rapid solution - 1 day after
  - Final solution - 12 days after
- High-rate analysis (>1Hz)
  - data from network HEPOS - 1Hz

The Scheme

The core tool/software is `Bernese GNSS Software v5.4`(**bernese**).
Integration with

- **PostgreSQL** database,
- **Python** module (product/data downloading, pre-processing, driving cron jobs, etc) https://github.com/DSOlab/autobern
- **Time-series** analysis (integrated in routine processing on regular intervals)
- **Strain Rates** via StrainTool (on user demand)



Switch to IGS20

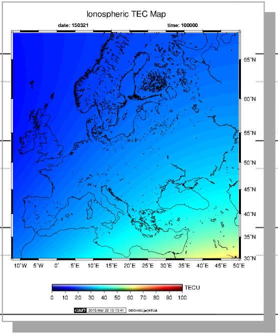
⇒ New release Bernese GNSS Software v5.4

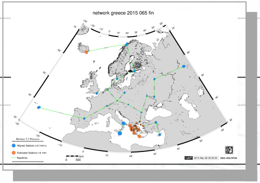
New values on processing options:

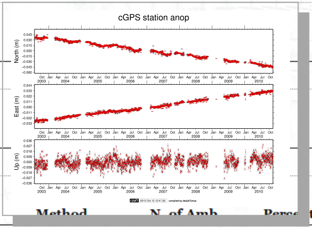
- Reference frame `IGS20`
- Oceans loading corrections (`FES2014`)
- `VMF3` for tropospheric modeling
- `ATX` file: `igs20.atx`
- Long filename of products according to new IGS convention

⇒ ⇒ Reprocess all available data from late '90s up to now

Results & Output







4. Solution Indetifiers

Array of Objects

expand

5. PCF Variables

Array of Objects

expand

6. Saved products

Array of Objects

expand

7. Warnings

Array of Objects

expand

8. Ambiguity Resolution SUmmary

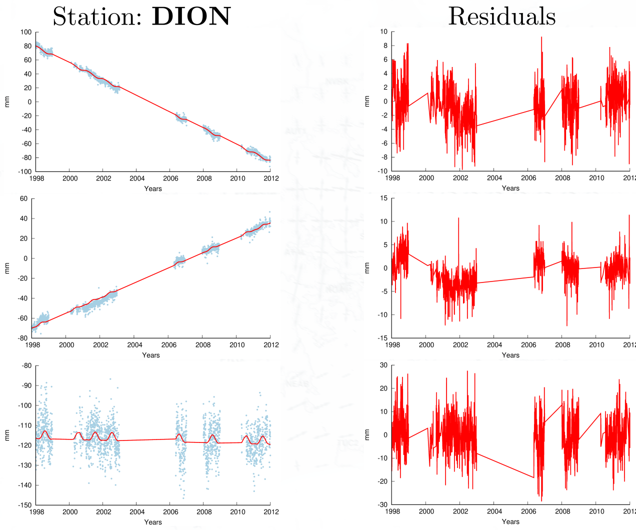
Array of Objects

Baseline	sta1	sta2	length (km)	Method	N. of Amb	Percentance	Satellite system
AUKL	AUT1	KLOK	139.7	pbnl	74	54.1	GPS
AULE	AUT1	LEMN	199.6	pbnl	60	55	GPS
KCTL	KATC	TILO	59	pbnl	50	90	GPS
KLRL	KLOK	RLSO	174.2	pbnl	74	41.9	GPS

Coordinate estimates - Time series analysis

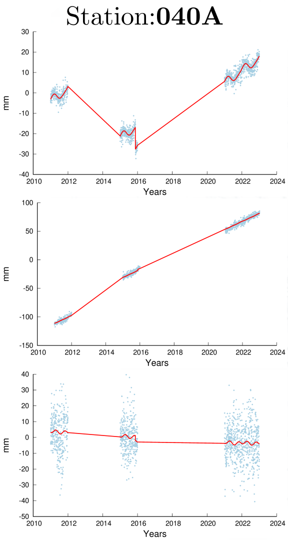
We analyze time-series using Hector Package (**Bos2012**), to estimate:

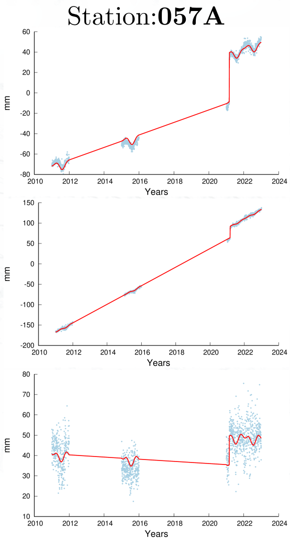
- tectonic velocities (linear trends),
- offsets/jumps due to miscellaneous reasons (e.g. instrumentation changes, earthquakes, etc); note that this step requires a-priori knowledge of such events (log-files, NOA earthquake catalogue)
- harmonics signals,
- velocity changes (e.g. inflation of Santorini isl.),
- post-seismic decay (still under development)



Earthquakes, co-seismic offsets

date	Δn	Δe (mm)	Δu
Station: 040A			
26.01.2014	-43.1	25.2	2.2
17.11.2015	-10.8	1.7	3.1
Station: 057A			
03.03.2021	47.9	27.6	14.0



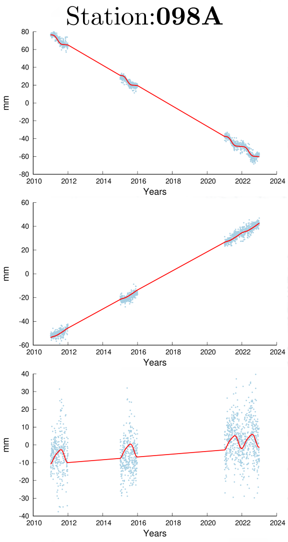


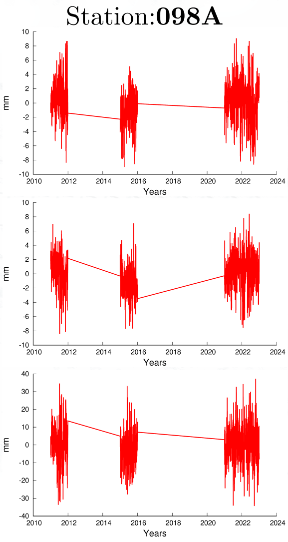
Harmonic signals

$$\sum_{i=0}^{n_F} s_i sin(\omega_i t) + c_i cos(\omega_i t)$$

Results:

```
-----------------------------
trend: -11.398 +/- 0.086 mm/year
-----------------------------
cos yearly : 1.428 +/- 0.255 mm
sin yearly : 1.804 +/- 0.280 mm
Amp yearly : 2.316 +/- 0.266 mm
Pha yearly : 51.629 degrees
-----------------------------
cos hyearly : -0.418 +/- 0.206 mm
sin hyearly : -0.128 +/- 0.217 mm
Amp hyearly : 0.493 +/- 0.195 mm
Pha hyearly : -162.958 degrees
-----------------------------
```

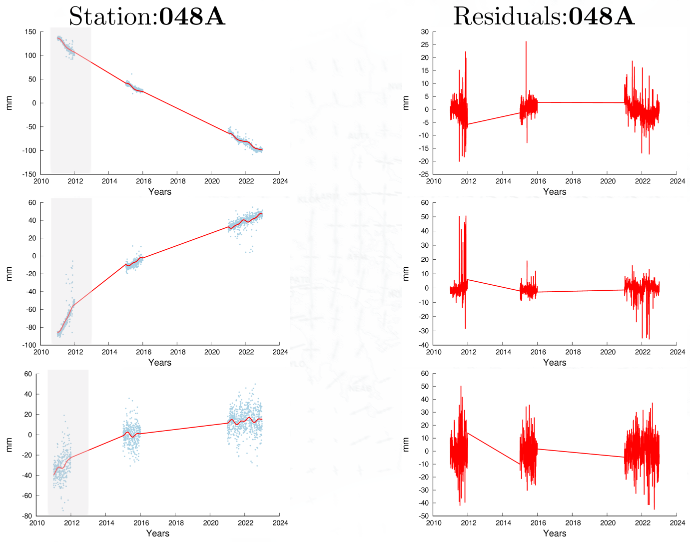




Specific case - Santorini

Estimate velocity changes in Santorini station:

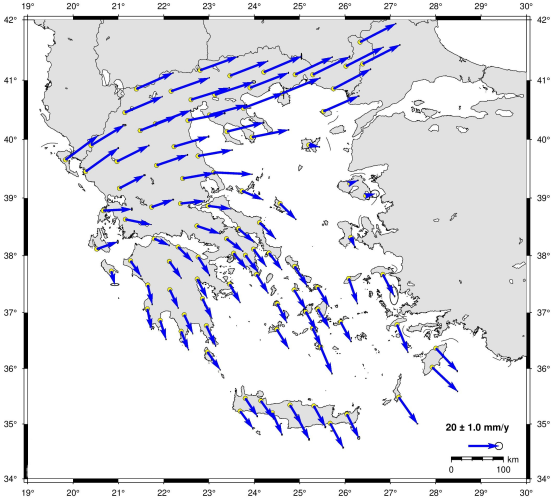
comp	< ∼2013 (mm/yr)	> ∼ 2013
north	-30.1	-17.5
east	31.3	7.0
up	17.2	2.1



Velocity Field of HEPOS Network

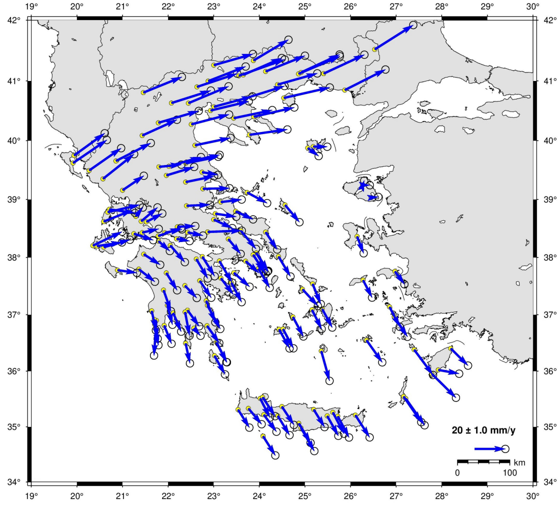
Estimated velocities (IGb14) vary:

comp	min	max
	(mm/yr)	
north	-17.9	15.5
east	2.4	26.0
up	-5.5	4.7

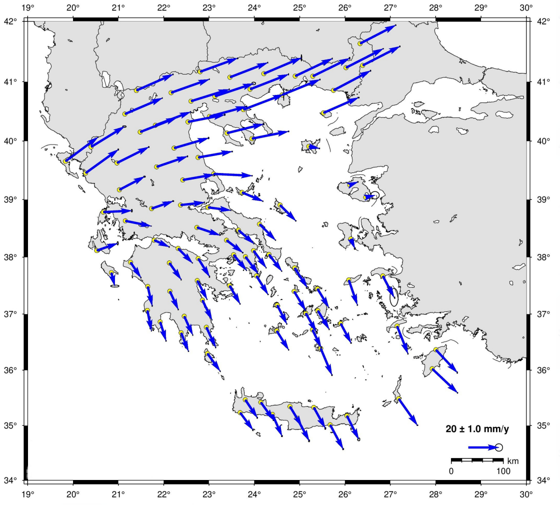


Compare velocity field of different networks

DSO_GRC_2023

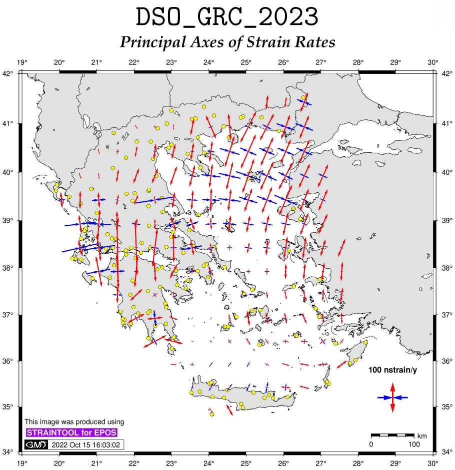


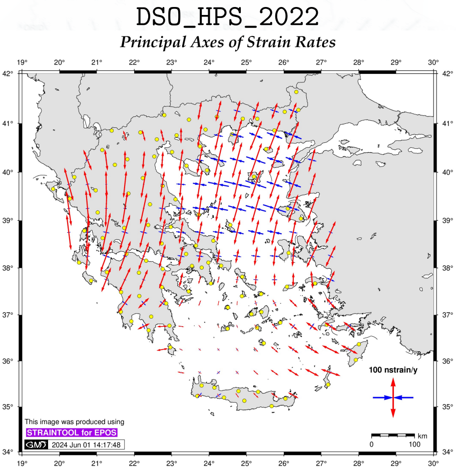
DSO_HPS_2022



Strain rates

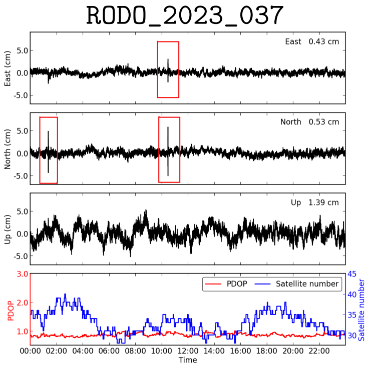
StrainTool software used to estimate strain tensor parameters (**straintool**)

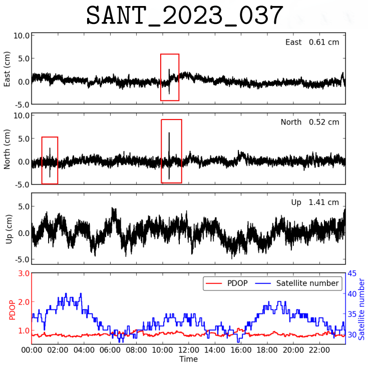


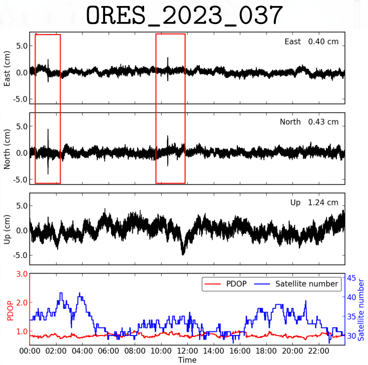


PPP processing for High-rate data

effect from Turkey earthquake (06.02.2023)



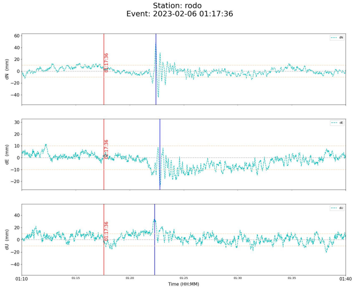


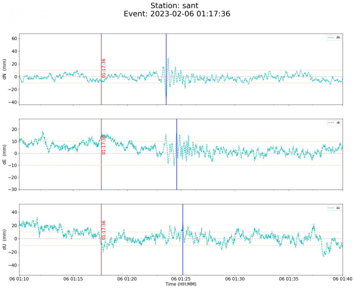


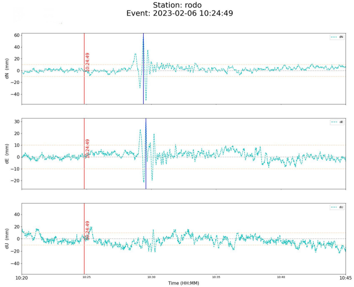
PRIDE Lab
Positioning Racers to Image & Decipher the Earth

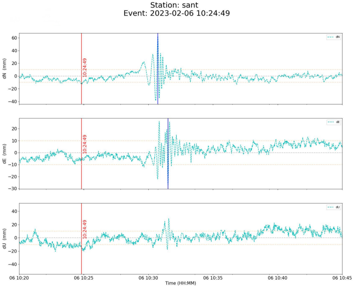
- 1Hz data available, ∼50 stations from Metrica SmartNET.
- PPP-AR analysis for 3 days (DOY: 036, 037, 038)

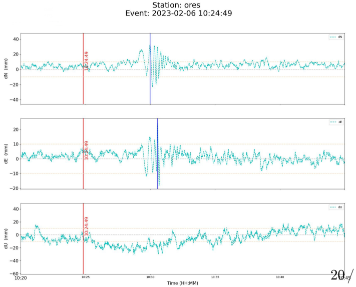
Focus on stations











Discussion / Conclusions

- Greece is located in a complex tectonic background with many changes in the kinematics of the area.
- Routine processing and monitoring are very important and revealing for Greece; products are requested by and disseminated to a wide range of Geoscientists.
- A dense velocity field for accurate estimation of ground/tectonic motions in the region will help to develop a stable local reference frame and the connection of the region with the global and European reference systems.
- The continuous monitoring of the networks gives useful results for the effect of strong earthquakes or other “abrupt” phenomena; small, dense networks are of great help (even with instrumentation of non-geodetic accuracy).
- DSO is upgrading its scientific output and processing activity via new datasets and real time processing.
- Our plan: grow up our team and be more involved to the GNSS/EUREF community.

Thank you for your attention!

References I